

Building Legal RAG: what each component contributes

Aleksandr Loginov, Yulia Malakhova

1. Research Context

RAG grounds language-model answers in external documents, but legal QA requires a pipeline to satisfy two goals at once: retrieve enough relevant evidence to produce a correct answer, while keeping source-page citations precise and complete. This is challenging because legal PDFs may lack machine-readable text, evidence can appear at different textual scales, and questions may depend on either exact legal references or semantic similarity. Moreover, retrieving more evidence can improve coverage while also introducing irrelevant source pages. We therefore examine how design choices in document processing, retrieval, and attribution can improve answer quality without sacrificing citation quality.

2. Prior Work & Study Focus

Retrieval-augmented generation (RAG) conditions language-model answers on retrieved documents [1]. Legal retrieval must balance precision and coverage: LegalBench-RAG targets minimal relevant passages [2], while regulatory QA studies show that retrieving all supporting evidence becomes harder when it spans multiple passages [3]. In an end-to-end legal benchmark, retrieval-model choice had a larger effect on answer correctness than generator choice [4].

Evaluation also extends beyond correctness. Legal QA research examines whether answers are supported by the supplied context [5], while ALCE evaluates answer correctness and citation quality separately [6]. Together, these studies motivate evaluating both the evidence available to the generator and the sources ultimately cited.

Our study complements these approaches by holding the generator fixed and examining a broader sequence of pipeline choices, from document processing to final page selection.

3. Research question

With the answer generator fixed, which pipeline additions improve answer quality and page-level citation precision and recall, and where do these objectives conflict?

4. Our Contribution

We build a Legal RAG pipeline and provide an empirical comparison of seven cumulative configurations on 30 documents and 100 questions. By adding components sequentially while keeping the answer generator fixed, we measure changes in answer quality and page-level citation precision and recall across stages of document processing, retrieval, reranking, answer processing, and citation selection.

5. Hypothesis

H1 – Retrieval

Combining multiple text granularities, semantic and lexical retrieval, and relevance reranking will improve answer quality and evidence retrieval compared with a simple semantic-retrieval baseline.

H2 – Robustness to scanned documents

Recovering text from scan-like legal documents will reduce the performance loss caused by the absence of a machine-readable text layer.

H3 – Citation attribution

Selecting source pages after answer generation will produce a more focused citation set without substantially reducing the coverage of relevant evidence.

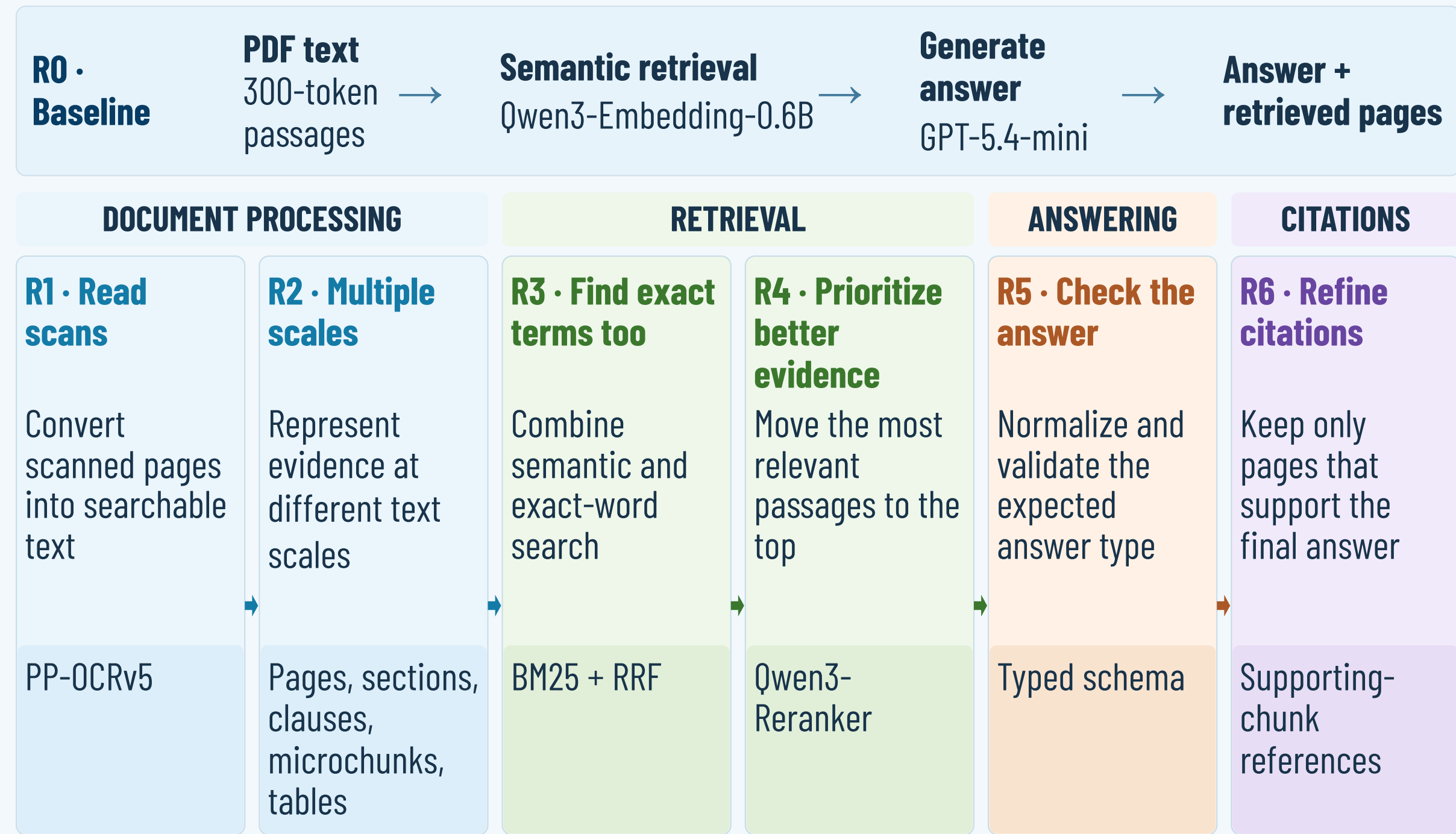
6. Data and Methods

Data

30 DIFC legal PDFs (590 pages), 100 questions, including 70 structured-answer and 30 free-text questions. Reference source pages are available for 95 questions. For the scan-robustness test, 10 of the 30 documents (107 pages) were converted into synthetic scans: each page was rasterised as an image and its machine-readable text layer was removed.

Pipeline

Starting from a simple baseline, we add one capability at a time. Each variant includes all previous additions.



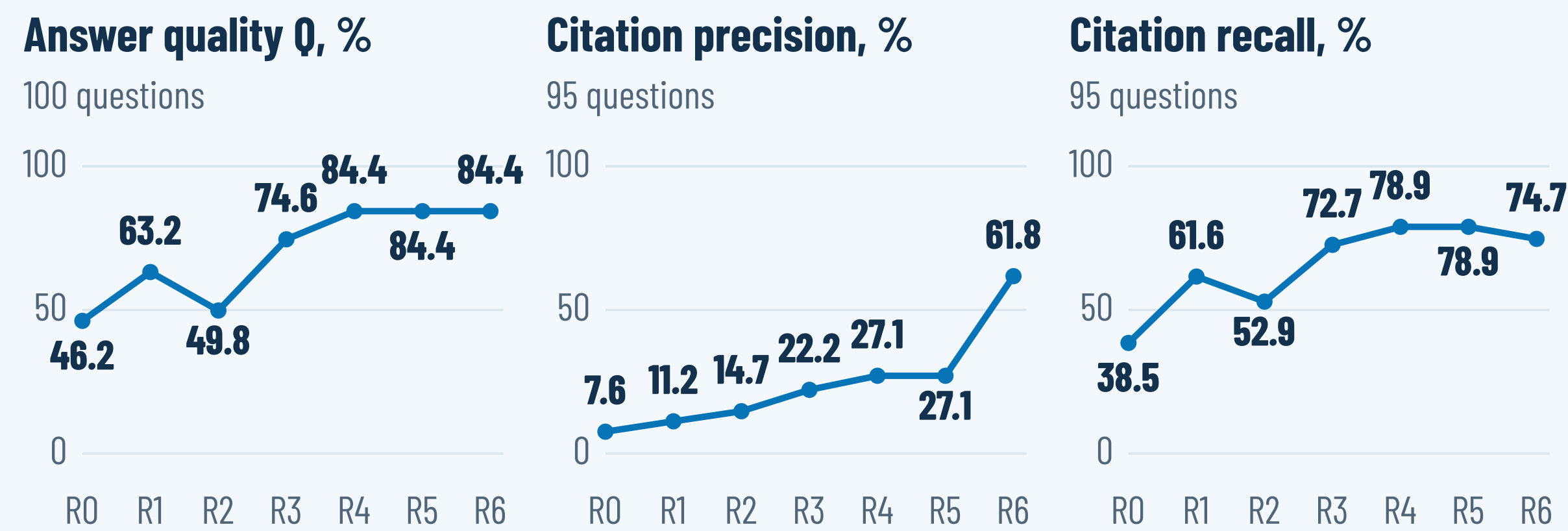
Answer generation: A fixed GPT-5.4-mini configuration is used across variants, so changes in performance reflect modifications to the surrounding pipeline rather than a change of the generating LLM.

7. Evaluation

A correct answer does not necessarily imply good citations. A system may produce the right answer while citing irrelevant pages, or it may cite only part of the evidence required to support the answer. We therefore evaluate answer quality and citation quality separately.

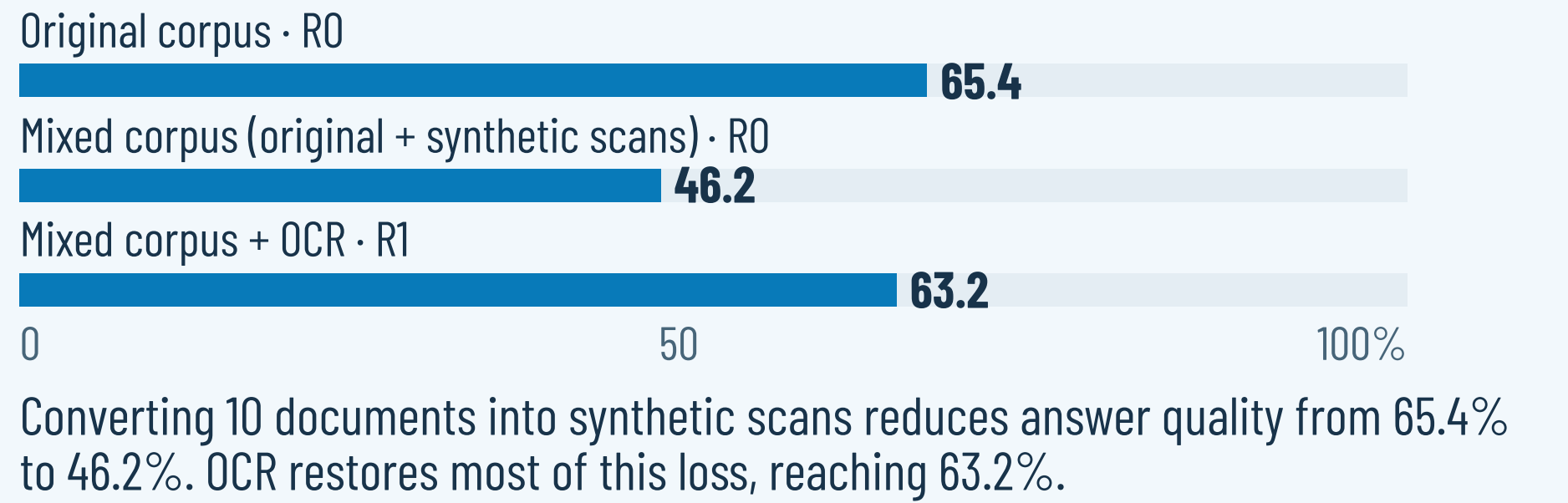
Answer quality	Citation precision	Citation recall
<i>Is the generated answer correct?</i>	<i>Are the cited pages actually relevant?</i>	<i>Did the system cite all required evidence?</i>
Q = 0.7 × structured score + 0.3 × free-text score	Share of cited pages that support the answer.	Share of annotated supporting pages that were cited.

8. Results: Pipeline performance



9. Results: OCR robustness

OCR recovers performance on scanned documents



10. Results: Hypothesis evaluation

- H1 – Retrieval & representation Partially supported.** Multi-scale representations alone reduce Q, but adding lexical retrieval and reranking substantially improves answer quality and source-page recall.
- H2 – Scan robustness Supported.** OCR restores most of the answer-quality loss caused by removing the machine-readable text layer.
- H3 – Citation selection Supported with a trade-off.** Precision rises from 27.1% to 61.8%, while recall decreases moderately from 78.9% to 74.7%.

11. Conclusion and takeaways

Legal RAG should be designed as a multi-stage pipeline: OCR improves robustness to unreadable PDFs, hybrid retrieval and reranking improve answer quality, and post-answer citation selection improves citation precision. However, citation selection introduces a precision-recall trade-off, so answer quality and citation quality should be optimized and evaluated separately.

- 1 · Ensure document robustness** OCR provides robustness when machine-readable text is unavailable. It recovered most of the performance loss introduced by synthetic scans.
- 2 · Optimize retrieval** Hybrid retrieval and reranking produced the largest observed gains in answer quality. Additional representation complexity did not consistently improve performance.
- 3 · Optimize citations separately** Post-answer citation selection substantially increased precision, with a small reduction in recall.

Unexpected outcome: Multi-scale representations did not improve performance; Q decreased from 63.2% to 49.8% after R2.

12. Limitations

one DIFC corpus and 100 questions; one run per condition with no significance testing; LLM-based evaluation of free-text answers.

13. References

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Code, protocol & results
github.com/SteinVR/Trends-in-NLP-Poster